

ULTRON 2.7 Phase 13 Execution Report

Beta-ready local prototype

Project	ULTRON 2.7
Phase	Phase 13: Beta-Ready Local Prototype
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 13 turns ULTRON 2.7 into a beta-ready local Windows prototype. The project now has setup scripts, dependency and provider checks, a config wizard, a one-command launcher, diagnostics, a no-microphone demo, and CI-style verification.

The safety architecture is unchanged. Operator tooling does not grant raw shell access or bypass schema validation, policy gates, executor restrictions, or audit logging.

Phase 13 Scope

- Add Windows setup and local dependency checks.
- Add STT/TTS/wake/VAD provider health checks.
- Add a config wizard for workspace, safe roots, app aliases, voice providers, and dry-run mode.
- Add a launcher that starts the backend and UI together and handles port conflicts.
- Add a diagnostics API and diagnostics page.
- Add a no-microphone demo flow using typed commands and mock voice.
- Add a CI-style verification script.
- Update README, architecture docs, install guide, run guide, troubleshooting guide, and safety guide.

What Was Implemented

Area	File(s)	Result
Diagnostics	src/ultron27/diagnostics.py, web_server.py	Adds /api/diagnostics with backend, brain, runtime path, voice provider, safe-root, dependency, and recent-error readiness data.
Launcher	scripts/launch_ultron.py	Starts backend and UI together, chooses an open port, and prints URLs.
Setup	setup_ultron_windows.ps1, check_dependencies.py	Creates a Windows venv, installs the project, writes safe defaults, and reports missing dependencies.

Area	File(s)	Result
Config	scripts/config_wizard.py	Generates ultron.config.json with workspace, safe roots, app aliases, voice providers, and dry-run defaults.
Demo	scripts/demo_phase13.py	Runs typed command, note creation, confirmation pause, and mock voice demos without a microphone.
Verification	scripts/verify_phase13.py	Runs tests, dataset evaluation, safety regression, and API smoke checks.
UI	web/diagnostics.html, diagnostics.js	Adds a readable diagnostics page linked from the main interface.

Beta Operator Workflow

Need	Command	Expected Outcome
Install	.\scripts\setup_ultron_windows.ps1	Creates .venv, installs package, writes config, checks dependencies.
Launch	python scripts\launch_ultron.py	Starts backend and UI, then prints URLs.
Diagnose	Open /diagnostics	Shows backend, brain, voice, memory, audit, safe roots, and recent errors.
Demo	python scripts\demo_phase13.py	Runs typed and mock voice demos without a microphone.
Verify	python scripts\verify_phase13.py	Runs tests, dataset, safety, and API smoke checks.

Verification Results

Check	Command	Result
Unit tests	python -m unittest discover -s tests	71 tests passed.
Compile check	python -m py_compile Phase 13 modules/scripts	Python modules and scripts compiled successfully.
Frontend syntax	node --check web/app.js and diagnostics.js	Frontend JavaScript parsed successfully.
Dependency check	python scripts/check_dependencies.py --json	Required paths and packages were available.
Provider check	python scripts/check_providers.py --json	Text/browser providers active with no warnings.
Demo flow	python scripts/demo_phase13.py	Typed, note creation, confirmation, and mock voice demos completed.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Total 285; intent 0.9965; tool 0.9965; arguments 0.9579; confirmation 1.0.
Safety regression	python scripts/evaluate_safety_regression.py	362 cases; tool accuracy 1.0; policy action accuracy 1.0.

Check	Command	Result
CI-style verifier	python scripts/verify_phase13.py	Passed tests, dataset, safety, and API smoke checks.

Safety Impact

Phase 13 does not add a new execution authority. Diagnostics are read-only, the launcher only starts the existing local web server, the demo uses typed and mock voice inputs, and the verifier exercises public API routes.

Recommended Next Step

- Package a signed Windows shortcut or small desktop launcher around scripts/launch_ultron.py.
- Add a first-run checklist in diagnostics for optional local voice model downloads.
- Add exportable demo logs for presentations and beta feedback.